

1M1B

CampusEnergy Twin

AI FOR SUSTAINABILITY

CampusEnergy Twin

1M1B AI for Sustainability Internship · IBM SkillsBuild &
AICTE

Primary SDG 7 Affordable and
Clean Energy

Presented by [Your Name] ·
[College Name]

[Program / Branch]

Campus energy is hard to see clearly.

Fragmented data hides the patterns behind cost, comfort, and waste.



Data overload

Campus systems generate plenty of data, but it lives in disconnected places.

- Meters, buildings, and schedules stay siloed
- Teams spend time finding signals



Limited visibility

Without a shared view, energy usage is difficult to compare, explain, or act on.

- Usage varies across buildings
- Schedules make context harder to read



Hidden waste

Unusual consumption can remain invisible until it becomes expensive, disruptive, or urgent.

- Anomalies are difficult to spot early
- Reactive fixes replace informed action

SUSTAINABILITY ALIGNMENT

SDG Alignment

Turning campus energy intelligence into measurable climate action.

UNITED NATIONS GOAL



7

Affordable and Clean Energy



Smart decisions for a lower-carbon campus.

How the twin advances SDG 7



Energy efficiency improvement

Identify where smarter operations can reduce waste.



Renewable energy planning support

Connect campus demand with clean-energy opportunities.



AI-based energy decisions

Translate live insight into confident action.



A smarter campus ecosystem starts with clean energy visibility.

AI-POWERED DIGITAL CAMPUS

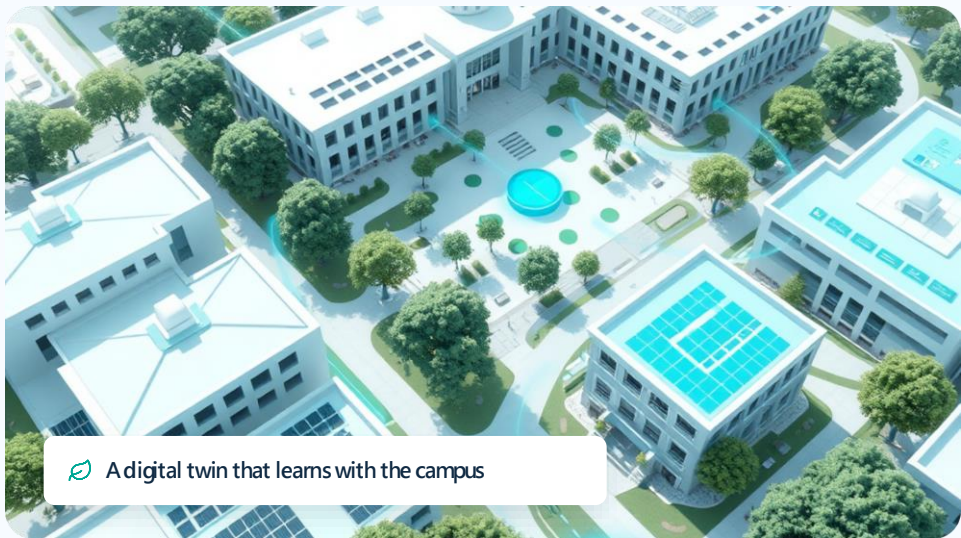
Our Solution

Turn campus energy data into a living system for smarter, more sustainable decisions.

From signal to strategy

- 01 Monitor performance**
See how every campus zone is using energy.
- 02 Predict demand**
Anticipate future patterns before they become peaks.
- 03 Detect and simulate**
Spot unusual consumption and test sustainability strategies.

→ One connected workflow for action



A digital twin that learns with the campus

How Does CampusEnergy Twin Get Data?

From fast-start prototype inputs to a connected, validated campus data layer.



PROTOTYPE TODAY

Quick-start inputs



CSV or manual uploads



Demo or synthetic prototype data

Start modeling before every live feed is available.



CONNECTED TOMORROW

Live campus sources



Smart meter and IoT integration



API connections to campus systems

Scale from isolated sources to an always-on ecosystem.



Every source is validated into one common data layer — the foundation for the digital twin.



AI-POWERED ENERGY INTELLIGENCE

From campus data to energy insights

A smarter operating layer for forecasting, detection, and action.



Forecast demand

Anticipate future energy needs before they become operational pressure.



Detect anomalies

Surface unusual patterns that can signal waste, faults, or changing use.



Recommend action

Turn live signals into clear, data-driven decisions for campus teams.



Campus data → energy insights

Observe · understand · improve

Intelligence in the loop



DIGITAL CAMPUS INTELLIGENCE

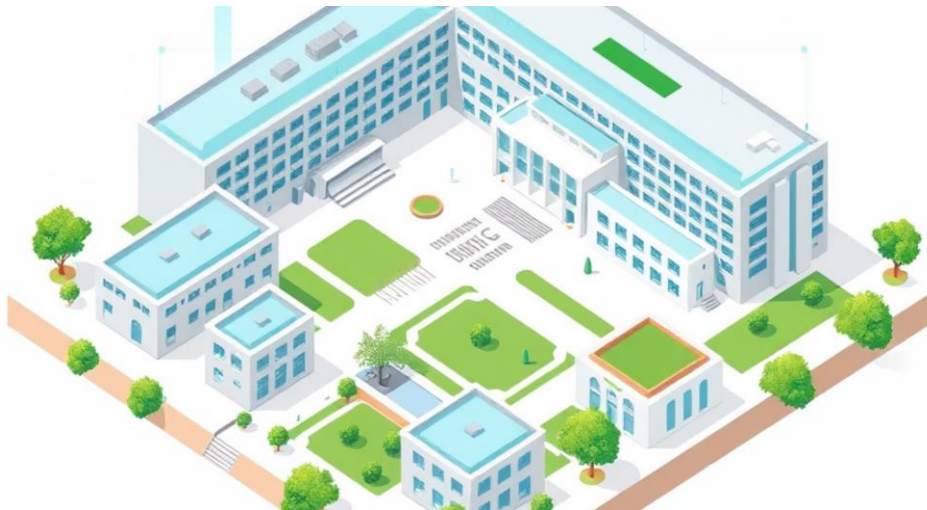
Digital Campus Twin & What-If Simulation

Move from today's campus reality to a simulated, lower-impact operating state.

Campus twin view

One connected model for buildings, systems, and schedules

Live campus model



● Built environment ● Energy systems ● Operational schedules

What-if scenarios



LED retrofit

Test lighting upgrades across blocks



AC optimization

Explore comfort and operating trade-offs



Schedule shift

Model demand around campus activity

Illustrative comparison

See the direction of change before committing to action.



Energy

Current → simulated



Cost

Current → simulated



CO₂e

Current → simulated

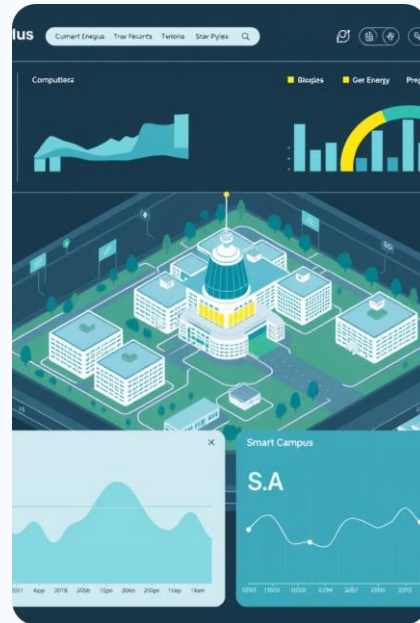
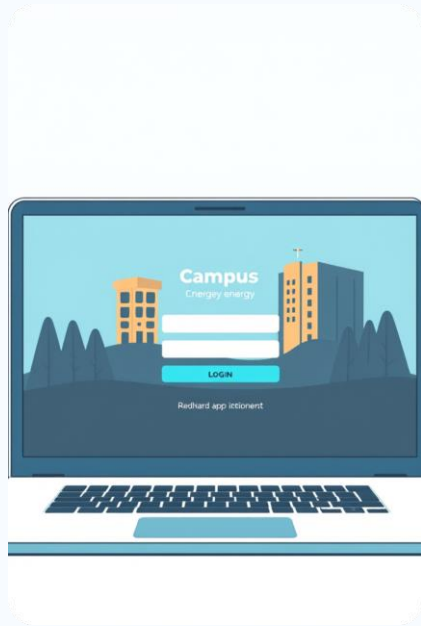
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APPLICATION PROTOTYPE

One connected view of campus intelligence

CAMPUSENERGY TWIN

From login to digital twin: roles predictions, and facility signals experience.



Prototype screens connect the full campus operating journey

SMART CAMPUS ECOSYSTEM

BUSINESS MODEL · EXPECTED IMPACT

From campus data to durable value

A recurring intelligence layer connects institutional revenue with more visible, sustainable campus planning.

Institutional subscriptions

Recurring access

Licensing creates a repeatable foundation for campus-wide adoption.



Visibility & anomaly response

Earlier awareness

A shared view helps teams spot unusual patterns before they become costly surprises.



AI analytics & reporting

Actionable insight

Clear reporting turns complex energy signals into decisions leaders can use.



Campus planning

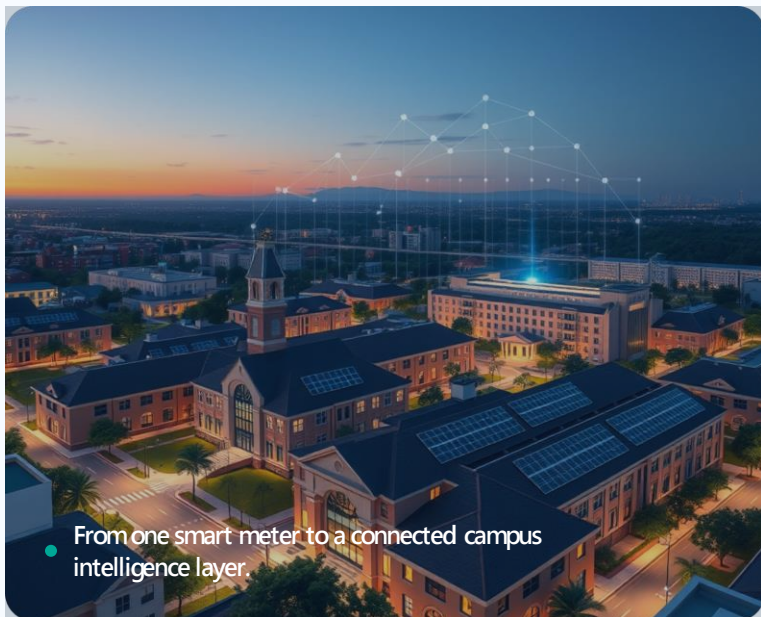
Sustainable direction

Long-term visibility supports practical, resilient choices across the campus ecosystem.



Build trust into every campus decision.

A responsible foundation today makes intelligence scalable tomorrow.



From one smart meter to a connected campus intelligence layer.



Responsible by design

Transparency, privacy, oversight, and clear uncertainty communication.



Trust the data boundary

The prototype uses demo data unless real campus data is intentionally uploaded.



Scale with purpose

Smart meters can grow into multi-campus intelligence.

TAKEAWAY

The future is not just smarter energy—it is more accountable energy.